github.com/puneet-chandna
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PUNEET CHANDNA

puneetchandna.com
puneetchandna@zohomail.in

Education

Bachelor of Technology (B.Tech) in Computer Science and Engineering (CSE) *Expected Graduation: 2026*
Vellore Institute of Technology, Chennai *CGPA: 7.42*
Senior Secondary (Class XII) *July 2022*
Vivekanand Public School

Skills

- **Programming Languages:** Python, C++, Java, SQL, JavaScript, Bash.
- **Tools & Frameworks:** Git, React, Next.js, Node.js, Express.js, RESTful APIs, FastAPI, Linux.
- **Databases & Cloud:** MongoDB, PostgreSQL, Supabase, Amazon Web Services (AWS), Google Cloud (GCP), Docker.
- **Core Concepts:** Data Structures & Algorithms, DBMS, CI/CD.

Certifications

Professional Cloud Architect Issued: Feb 2026
Google Cloud | Credly Profile: credly.com/users/puneet-chandna.ea5ad5b2

Experience

Apoliums Infotech India Pvt. Ltd. Dec 2025 – Jan 2026
Backend Engineering Intern
- Migrated legacy Node.js services to Golang (Gin) and optimized MySQL schemas, reducing latency and improving concurrent request handling.
- Designed and deployed RESTful backend services on GCP using Compute Engine and Cloud SQL, improving scalability and stability.

Research Intern May 2025 – July 2025
Centre for e-Automation Technologies (CeAT), VIT Chennai
- Developed a **CloudSim Plus** framework to evaluate **Hippopotamus Optimization (HO)** algorithm for VM placement in data centers.
- Enhanced simulation scalability by modularizing components, enabling reproducible testing across multiple datacenter configurations —**GitHub:** github.com/puneet-chandna/cloudsim-ho-research

Full Stack Developer Intern Dec 2024 – Feb 2025
Daira Edtech Pvt Limited
- **Architected the Activity Management System** using Node.js and MongoDB, engineering RBAC with secure **JWT** flows to facilitate project lifecycles for faculty and external reviewers.
- **Enhanced system reliability** and Student Progress tracking by refactoring legacy logic and implementing rigorous server-side testing, significantly reducing API errors and downtime.

Web Development Intern Sept 2024 – Oct 2024
Indian Institute of Technology Bombay (IIT Bombay)
- Reduced JSON payload size by 90% for low-bandwidth environments by replacing text messages with numerical status codes.
- Enhanced backend robustness and API reliability by implementing Joi schema validation for the SaaS platform.

Projects

Real-Time Dealer Gamma Exposure (GEX) Analysis for 0DTE SPX Options Dec 2025 – Apr 2026
- Built a real-time 0DTE SPX dealer GEX engine in FastAPI, pricing 1000+ option contracts in <1 ms using vectorized Black–Scholes–Merton Greeks with expiry normalization and contract-level OI/IV filtering.
- Built a Next.js live dashboard for strike-level GEX, cumulative exposure, zero-gamma levels, and Charm/Vanna signals, served through cached REST APIs, 5 s WebSocket updates, and PostgreSQL session storage.
- Validated the relationship between net GEX and realized volatility using Welch's t-test, finding significantly higher volatility below $-\$1$B net GEX while sustaining sub-200 ms API latency and 99.5% WebSocket uptime.

Technologies: Python 3.13, FastAPI, NumPy, SciPy, Pydantic, PostgreSQL, Next.js, TypeScript , Zustand, WebSockets, Recharts
Github: github.com/puneet-chandna/0DTE-dealer-gamma

Key-Adaptive Parallel AES Encryption Engine Jan 2025 – Apr 2025
- Reduced encryption time by 51% on large payloads by engineering parallel AES-128 encryption using Python's `ProcessPoolExecutor`.
- Increased throughput by 21.9% (362 KB/s vs. 297 KB/s) through a multi-core CTR-mode pipeline optimized across CPU cores.
- Designed SHA-512–seeded dynamic S-box generation with 49.95% avalanche effect, preserving AES-grade diffusion properties.

Technologies: Python, Multiprocessing, SHA-512, PBKDF2-HMAC-SHA256, Matplotlib, NumPy

Water Brakes Aug 2024 – Nov 2024
- Developed a Streamlit app analyzing contour maps to recommend precise swale and trench locations for water management.
- Enabled data-driven groundwater recharge decisions by processing UTM data to generate interactive Plotly visualizations.

*Technologies: Streamlit, Python, Plotly, Matplotlib | – **Live:** https://water-brakes.streamlit.app*

Positions of Responsibility

Outreach Lead, DAO Community:
- Grew community membership by 50% in 6 months through structured outreach, workshops, and collaborations with Web3 companies including Polygon and Huddle01.
- Secured partnerships for 15+ national hackathons and TOKEN2049 Singapore while leading a 35-member team to organize 5 large-scale hackathons, including Defy24, generating ₹5.5L+ in sponsorships and ₹3.5L+ in profit.

Blockchain Lead, Hackclub:
- Designed blockchain workshops, structured learning roadmaps, and mentorship programs to accelerate members' Web3 skills.